30 DAYS PYTHON MICRO COURSE

BY SATISH DHAWALE

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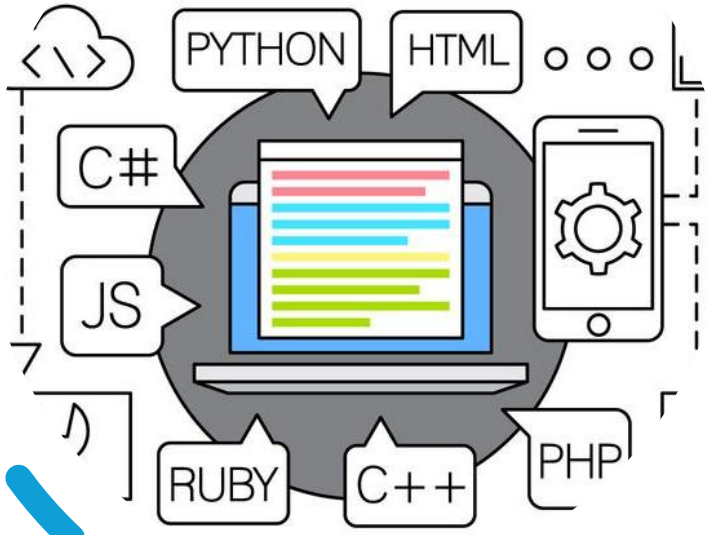
Day-1-Introduction to Python & Installation

What is a Programming Language?

A **programming language** is a formal set of instructions that tells a computer what to do. These instructions follow a specific syntax and rules so that the computer can interpret and execute them.



Layman Terms

Computers do not understand human language directly; they understand **binary (0s and 1s)**. Programming languages bridge this gap by allowing humans to write code in a readable format, which is then translated into machine code.

Think of a programming language like giving directions to a taxi driver:

You tell the driver step-by-step where to go (**code/instructions**).

The driver (**computer**) follows exactly what you say.

If your directions are clear, you reach the right place (**desired output**).

If your directions are wrong or incomplete, the driver might take you somewhere else (**errors or wrong output**).

HTML
JAVAFX
C++
ASSEMBLER
GROOVY
DELPHI
VBASIC
SEM
PASCAL
DELPHI
PERL
ACTIONSCRIPT
JAVASCRIPT
PROGRAMMING
PYTHON
BASIC
ASSEMBLER
JAVASCRIPT
RUBY
MYSQL
XML
BASIC
PROGRAMMING
JAVA
C++
PYTHON
PROGRAMMING
PHP
SQL

What is Python?

Python is a **high-level, interpreted, object-oriented programming language** created by **Guido van Rossum** and first released in **1991**.

Python is just like learning a new language — but easier than English grammar.
It lets computers understand your instructions.

Why Python is Super Easy for Non-Technical People

EASY TO READ

LOOKS LIKE ENGLISH

VERY FEW RULES

HUGE COMMUNITY SUPPORT

Why Python for Data Analysts

- Simple data cleaning
- Works with CSV/Excel
- Analyze large datasets
- Automation (emails, reports)
- Data visualization

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Real-Life Use Cases for Data Analysts

1. Cleaning messy Excel files
2. Removing duplicates
3. Automating daily reports
4. Creating dashboards
5. Predicting trends (basic ML later)

History of Python

- Created by Guido van Rossum
- Released in 1991
- Named after “Monty Python” comedy show
- Now world’s #1 used programming language

What Can You Build with Python?

- Websites
- Apps
- Games
- Automations
- Data analysis
- AI / Machine Learning

Career Perspective:

PYTHON DEVELOPER

DATA ANALYST

DATA SCIENTIST

AI/ML ENGINEER

AUTOMATION ENGINEER

Python Installation (Step-by-Step)

Download	Download from python.org
Install	Install Python 3.x
Check	Check “Add Python to PATH”
Install	Install VS Code

Step 1 – Download Python

- Go to the official website: https://www.python.org/
- Hover over **Downloads**.
- Choose your operating system (Windows/Mac/Linux).
- Download the latest **Stable Release** (e.g., Python 3.x.x).
- **Note:** Always check **"Add Python to PATH"** during installation (important for running Python from command line).

Step 2 – Install Python (Windows Example)

- Open the downloaded installer.
- Tick **"Add Python to PATH"**.
- Click **Install Now**.
- Once installed, open **Command Prompt** and type:
- python --version
- This should show the installed version.

Install Required Extensions in VS Code

- Python Extension by Microsoft
- Jupyter Extension
- Code Runner (Optional)

First Python Program

- print("Hello Data Analysts!")

Summary of Today's Lesson

- What is Python?
- Why Python for Data Analysts
- Download & Installation
- VS Code Setup
- First Program

Assignment for Students

1. Install Python
2. Install VS Code
3. Run 3 print programs
4. Take a screenshot of VS Code window

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